

Use It or Lose It?

Programmable money and the digital euro's interest rate

[Presenter name]

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Money with an expiry date gets spent, and fast

China, digital coupons, 2020

¥3

extra spending per ¥1 of subsidy
coupons valid for about one week

France, randomised prepaid cards

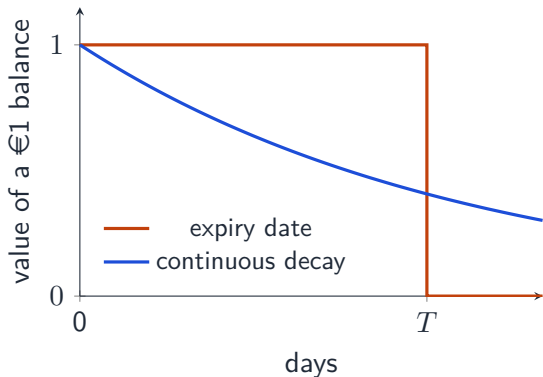
61% vs **23%**

one-month MPC, with vs without
a three-week expiry

A deadline turns a transfer that would be saved into spending *now*.

Xing, Zou, Yin, Wang & Li (AEJ: Macro 2023); Ding, Jiang, Msall & Notowidigdo (2024); Boehm, Fize & Jaravel (2024).

A digital euro could build the same pressure into money itself, by an expiry date or by continuous decay



- In a liquidity trap, money is hoarded and velocity falls
- A programmable currency can charge for holding it
- **Expiry:** the whole cost falls on day T
- **Decay (demurrage):** the same cost spread evenly over time

Which of the two should a central bank use?

To compare them, take a holder who searches for good uses of money and can always dump it at a loss

- A unit of money is worth V to its holder; a good purchase is worth 1
- Searching at intensity κ costs $\kappa^2/2a$; κ is the **velocity**
- A bad use (junk goods, selling at a discount) is always available at $\phi < 1$

Expiry at T ($\tau = T - t$ left)

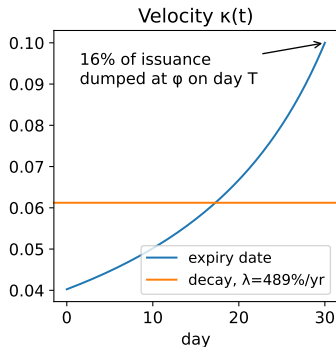
$$\frac{dV}{d\tau} = -rV + \frac{a}{2}(1 - V)^2, \quad V(0) = \phi$$

Decay at rate λ

$$(r + \lambda)V = \frac{a}{2}(1 - V)^2$$

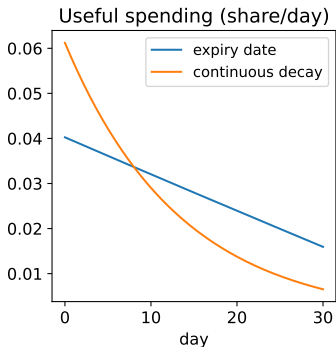
In both cases the holder spends at speed $\kappa^* = a(1 - V)$: the less money is worth holding, the faster it moves.

An expiry date pushes out more spending, but ends in a fire sale and splits money into vintages



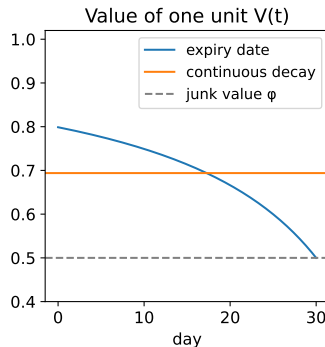
84% vs 73%

useful spending in 30 days



16%

dumped at ϕ on day T



0.50–0.80

value of an expiring unit, by age

Both schemes calibrated to the same average velocity over 30 days; $a = 0.2$, $\phi = 0.5$, $r = 4\%$.

Continuous decay keeps every euro equal, and it is nothing more than a negative interest rate

	Expiry	Decay
Fire sale at the deadline	3–55%	none
Value of one unit	by vintage	all equal
One euro = one euro?	no	yes
Applies to all money	no	yes

Decay at λ is the rate $i = -\lambda$.

Velocity follows

$$\kappa^* \approx \sqrt{2a(r + \lambda)}$$

the square-root rule of Baumol–Tobin money demand.

Expiry is a feature of a spending programme. Decay can be a feature of money.

As a rate, decay is far too weak to replicate a voucher: money can carry a rate, not a stimulus programme

Decay needed to match a voucher

160–1,200%

per year (90-day to 7-day vouchers)

Policy-sized rates, -0.5% to -3%

+6–32%

velocity

- Vouchers also work through new money, a salient deadline and minimum-spend thresholds
- None of these can be a permanent property of a currency

So the real question is the digital euro's interest rate, and the draft regulation fixes it at zero

“The digital euro shall not bear interest.”

Commission proposal for a digital euro regulation, Art. 16(8), June 2023

- Not positive, not negative, in any circumstances
- Paired with a holding limit, currently discussed at €1,500–3,000 per person
- Council and Parliament are negotiating the final text now

A zero-rate digital euro is cash without storage costs, which lifts the lower bound back to zero

Depositors can always switch to central-bank money, so banks cannot pay less than

$$r_D \geq \max_j \{i_j - c_j\} \quad (i_j: \text{rate}, c_j: \text{cost of holding})$$

	rate	holding cost	floor for deposit rates
Cash	0	storage, insurance $c > 0$	$-c$
Digital euro at zero	0	≈ 0	0

- 2019–22: the ECB went to -0.50% because hoarding banknotes is costly
- Next time, every holder can move up to the limit into a costless zero-rate asset
- Deposits leave banks exactly when negative rates already squeeze their margins

A negative digital-euro rate can win that space back, but no more while cash still exists

$$\text{floor} = \underbrace{\max\{0, -c\}}_{\text{zero rate}} = 0 \quad \longrightarrow \quad \underbrace{\max\{-\lambda, -c\}}_{\text{decay at } \lambda} \geq -c$$

- A zero rate **raises** the floor from $-c$ to 0
- Decay can bring it back down to $-c$, but **not below**: beyond that, people switch to banknotes
- Going deeper requires repricing cash itself, e.g. a crawling cash–reserve exchange rate

A flexible rate protects the policy space we have; it cannot create more.

The best design keeps zero as the default and lets balances above a threshold decay when policy rates turn negative

A contingent rate clause

$$i_{DE} = \begin{cases} 0 & \text{on all balances, when DFR} \geq 0 \\ 0 & \text{on balances up to } M_1 \\ \max\{\text{DFR}, -\bar{c}\} & \text{on balances above } M_1, \text{ when DFR} < 0 \end{cases}$$

- **Zero in normal times:** simple, trusted, never “money that melts”
- **Small balances never decay:** households and payments are protected
- **Decay, never expiry:** every euro stays equal to every other
- **Floored at $-\bar{c}$:** no reason to flee into banknotes

The digital euro should be able to bear interest: only as decay, and only when rates are negative

1. Deadlines make people spend, but an expiry date belongs to a spending programme, not to money
2. The only holding cost money can carry is continuous decay, which is simply a negative interest rate
3. Fixing that rate at zero turns the digital euro into costless cash and gives up policy space
4. A contingent, tiered rate keeps zero for normal times and protects that space when it matters

Backup

B1. Derivations

Holder's problem. Per unit of balance:

$$(r + \lambda)V = \max_{\kappa} \left\{ \kappa(1 - V) - \frac{\kappa^2}{2a} \right\} \Rightarrow \kappa^* = a(1 - V), \quad (r + \lambda)V = \frac{a}{2}(1 - V)^2.$$

Square-root rule. With $x = 1 - V$ small: $(r + \lambda)(1 - x) = \frac{a}{2}x^2 \Rightarrow x \approx \sqrt{2(r + \lambda)/a}$, so $\kappa^* \approx \sqrt{2a(r + \lambda)}$.

Expiry. With τ left, $dV/d\tau = -rV + \frac{a}{2}(1 - V)^2$, $V(0) = \phi$. If $\frac{a}{2}(1 - \phi)^2 > r\phi$, V rises above ϕ at once, so dumping happens only at T . Velocity $\kappa(\tau) = a(1 - V(\tau))$ rises to $a(1 - \phi)$: finite, but the unspent balance is spent all at once at T .

Secondary market. An expiring unit trades at $V(\tau)$; units of different ages trade at different prices.

Calibration. λ set so that decay velocity equals the average expiry velocity over $[0, T]$.

B2. Results by voucher length

T	Expiry				Decay (same average velocity)		
	useful	dumped	real waste	unit value	λ p.a.	useful	decayed
7 d	45%	55%	27%	0.50–0.63	1172%	41%	15%
14 d	66%	35%	17%	0.50–0.71	842%	58%	17%
30 d	84%	16%	8%	0.50–0.80	489%	73%	16%
90 d	97%	3%	2%	0.50–0.91	161%	88%	10%

Decay λ (p.a.)	0.5%	1%	2%	3%	5%
Velocity multiplier	1.06	1.12	1.22	1.32	1.49

Real waste = $(1 - \phi) \times \text{dumped}$ is a resource loss; decayed balances are a transfer to the central bank.

B3. Sensitivity ($T = 30$ days)

ϕ	a	useful	dumped	real waste	unit value	λ p.a.	useful	decayed
0.3	0.2	90%	10%	7%	0.30–0.77	833%	73%	22%
0.5	0.2	84%	16%	8%	0.50–0.80	489%	73%	16%
0.7	0.2	72%	28%	8%	0.70–0.84	211%	68%	9%
0.5	0.1	67%	33%	16%	0.50–0.71	403%	59%	17%
0.5	0.4	94%	6%	3%	0.50–0.87	506%	83%	12%

Across parameters: expiry front-loads more, always ends in a fire sale and always prices units by age; matching decay is always in the hundreds of percent per year.

Not modelled: general equilibrium; behavioural channels (salience, thresholds); spending after T (evidence mixed: Xing et al. 2023 vs Ding et al. 2024).

B4. References

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